TEST &



Fan Reliability Qualification: A Standardized Framework Across the Server Fan Supplier Base

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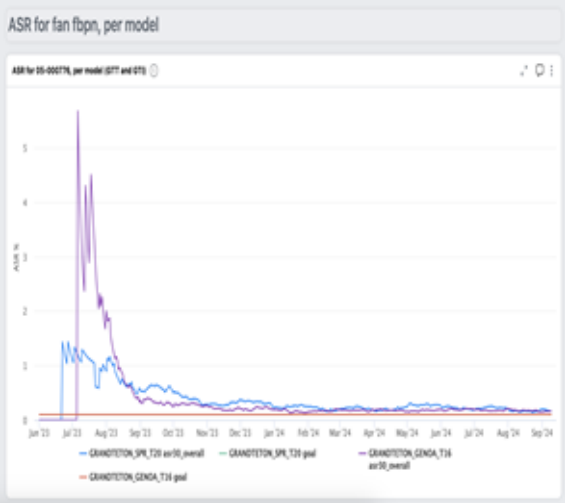
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OCP PLATINUM Member

Agenda

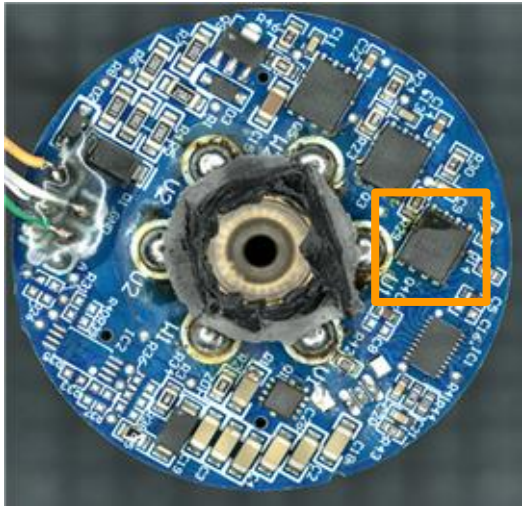
- Fan Reliability: Field Issues and Future Challenges
- Supplier Reliability Qualification: Wide Variation
- Solution: Standardized Fan Reliability Qualification
- Standardized Fan Reliability Qualification: Impact
- Conclusions and Future Work

Fan Reliability: Field Issues and Future Challenges



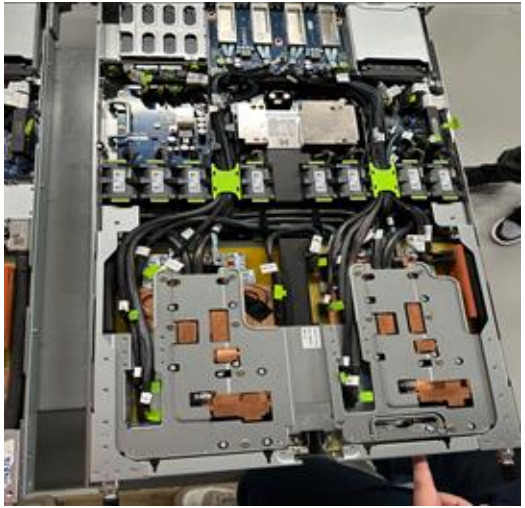
ASR's out of Spec

- *ELF Failures*
- *Similar trend across suppliers*



Myriad Failure Mode

- *Fan impeller cracks*
- *Electrical shorts*
- *MLCC capacitor cracks*
- *Solder defects*



The Future:

- *Non-Hot Swappable*
- *New Suppliers*

Comparative Assessment: Fan Supplier Reliability Qualification

1. Wide variation in supplier reliability testing, sample sizes and stress test profiles.

1. Limited reliability oversight into test procedures and results.

1. No “level-playing-field” for “apple-to-apple” evaluation of fan reliability across suppliers.

	Supplier A	Supplier B	Supplier C	Standardized requirement
High Temperature	85±2 °C 96 hrs · n = 5	70 °C 168 hrs · n not stated	Not Done	**85 °C · 7 days Vmax & Vmin + 1000 power cycles · n ≥ 22**
Low Temperature NEW TEST	–	–	–	**−40 °C · 7 days Vmax & Vmin + 1000 power cycles · n ≥ 22**
Humidity Exposure	75±2 °C · 85% RH 96 hrs · n = 5	60 °C · 90-95% RH 10 days · n = 3	60±2 °C · 90-95% RH 96 hrs · n not stated	**75 °C · 90-95% RH 10 days · n ≥ 22**
Thermal Cycling	−25 / +85 °C 2 cycles · 1 °C/min	Not Done	−10 / +70 °C 50 cycles · 1 °C/min	**−40 / +85 °C 300 cycles · 15 °C/min 30 min dwell · n ≥ 22**
Lifetime Demonstration NEW TEST	–	–	–	**> 90% reliability at 5 yrs, 35 °C n > 22**

meets the standard performed, but weaker not performed

Solution: Single Standardized Fan Reliability Qual. Procedure

- Developed standardized requirements across suppliers in a fan reliability requirements document
  - Address test profile gaps and a consistent set of tests across suppliers
  - Increased sample size requirements for higher confidence in results
  - Leverage IPC-9591 test processes
- Deployed standardized fan reliability requirements across all major fan suppliers
- Added continuous reliability monitoring (ORT) at suppliers to ensure consistent reliability in mass production

#	Test item	Sample size	Test condition	Change vs. prior practice
1	**High Temperature**	≥ 22	85 °C, 7 days at Vmax and Vmin, each followed by 1000 power on/off cycles	Duration 96–168 hrs → 7 days; power cycling added
2	**Low Temperature**	≥ 22	−40 °C (non-condensing), 7 days at Vmax and Vmin, each followed by 1000 power on/off cycles	**New test — no supplier ran a low-temperature profile**
3	**Humidity Exposure**	≥ 22	75 °C, 90–95% RH (non-condensing), nominal rated voltage, 10 days	Harmonises the 60–75 °C and 96 hrs–10 days spread
4	**Non-Operational Thermal Cycling**	≥ 22	−40 / +85 °C, 300 cycles with 30 min dwell, ramp 15 °C/min, checkpoints at 0 / 150 / 300	2–50 cycles → 300; ramp 1 °C/min → 15 °C/min
5	**Lifetime Demonstration**	> 22	> 90% reliability at 5 years, 35 °C ambient	**New test — no supplier demonstrated a lifetime target**

Impact of Standardizing Fan Reliability Qualification

1. Impeller Cracks in Thermal Cycling Tests

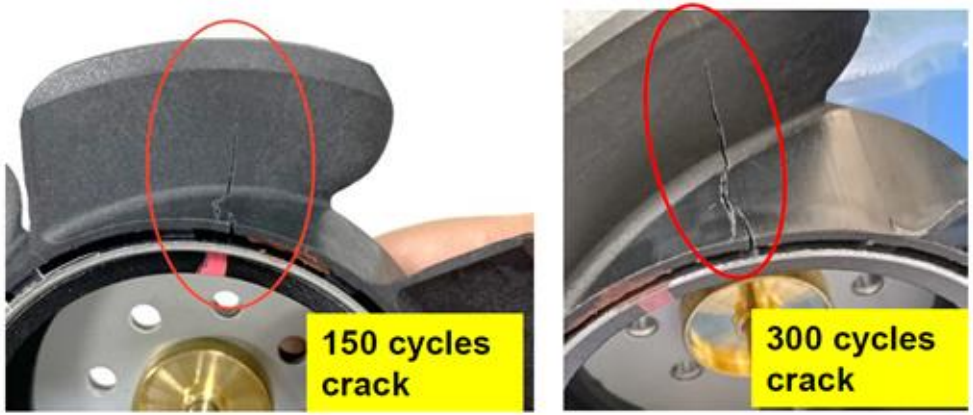
- **Root Cause:** Inconsistent impeller material specifications among suppliers.
- **Resolution:** Transitioned impeller material from Polytrimethylene Terephthalate (PPT) to Polyphenylene Sulfide (PPS).

2. Improved Fan Failure Rate in the Fleet

- Stable fan reliability for products deployed with new material requirements.

3. Availability gain for AI products

- Reduced server downtime due to fan repair



Acknowledgements

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  - Nikesh Tayi Dhar (Hardware Product Integrity Manager)

- Fan Suppliers
  - Sunon
  - Delta
  - AVC
  - Nidec

Call to Action

- Problem to solve: Fan suppliers face a different reliability qualification from every customer. One openly specified profile removes that duplication and raises the floor for everyone.
- Review and adopt the standardized qualification.
- Co-develop an in-system fan stress and diagnostic suite under OCP Test & Validation.
- Collaborate and Publish supplier qualification results to a common schema.

How to get involved

- Contribute to project and sub-project, monthly call, mailing list, wiki URL

Thank You!

